DevOps Lab — Commands Only (Exp 3–10)

Exp 1 & 2 = theory, no commands. Windows = PowerShell (not Git Bash). Replace <...> with your values.

Exp 3 — Git & GitHub

WINDOWS

```
winget install --id Git.Git -e
winget install --id Python.Python.3.12 -e
git --version
mkdir DevOps_Tools_Lab
cd DevOps_Tools_Lab
notepad matrix_multiplication.py
python matrix_multiplication.py
```

LINUX

```
sudo apt update && sudo apt install -y git python3
git --version
mkdir DevOps_Tools_Lab && cd DevOps_Tools_Lab
nano matrix_multiplication.py
python3 matrix_multiplication.py
```

matrix_multiplication.py

```
A = [[1, 2], [3, 4]]
B = [[5, 6], [7, 8]]
result = [[0, 0], [0, 0]]
for i in range(2):
    for j in range(2):
        for k in range(2):
            result[i][j] += A[i][k] * B[k][j]
print("Result:")
for row in result:
    print(row)
```

BOTH

```
git init
git config --global user.email "<email>"
git config --global user.name "<username>"
git add .
git commit -m "added matrix multiplication"
git branch -M main
git remote add origin https://github.com/<username>/DevOps_Tools_Lab.git
git push -u origin main        # Linux password = GitHub token
cd ..
git clone https://github.com/pragnagosula/Devops_Tools_Lab.git
cd Devops_Tools_Lab
dir                            # Linux: ls
```

Exp 4 — Docker + Jenkins install

WINDOWS (admin)

```
wsl --install
winget install --id Docker.DockerDesktop -e     # reboot, open Docker Desktop
winget install --id Oracle.JDK.21 -e
winget install --id Jenkins.Jenkins -e
java --version
Get-Service Jenkins
type "C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword"
```

LINUX

```
sudo apt update
sudo apt install -y docker.io fontconfig openjdk-21-jre
sudo systemctl enable --now docker
sudo usermod -aG docker $USER && newgrp docker
java --version
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/ | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update && sudo apt install -y jenkins
sudo systemctl enable --now jenkins
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

BOTH

```
docker --version
docker info
docker run hello-world
docker ps -a
# browser: http://localhost:8080 → paste password → Install suggested plugins → create admin
```

Exp 5 — Jenkins master–slave

```
Manage Jenkins → Nodes → New Node: Slave1 (Permanent) · executors 1 · label slave1
remote dir C:\Jenkins\slave (Linux ~/jenkins-slave) · Launch: connect it to the controller → Save → click Slave1, copy secret
```

WINDOWS (new terminal, keep open)

```
mkdir C:\Jenkins\slave
cd C:\Jenkins\slave
curl.exe -sO http://localhost:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://localhost:8080/ -secret <SECRET> -name Slave1 -webSocket -workDir "C:\Jenkins\slave"
```

LINUX (new terminal, keep open)

```
mkdir -p ~/jenkins-slave && cd ~/jenkins-slave
curl -sO http://localhost:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://localhost:8080/ -secret <SECRET> -name Slave1 -webSocket -workDir "$HOME/jenkins-slave"
```

```
New Item → Slave_function → Freestyle
☑ Restrict where this project can be run → slave1
Git URL: https://github.com/Sahithi-2408/jenkins_sample.git · Branch */main (else */master)
Build step → Windows: Execute Windows batch command · Linux: Execute shell:
echo Hello from Slave1
python app.py        # Linux: python3 app.py
Save → Build Now → Console Output
```

Exp 6 — Docker lifecycle (both OS)

```
docker pull ubuntu
docker images
docker run -it ubuntu bash
pwd
ls
whoami
exit
docker ps -a
docker start <id>
docker ps
docker exec -it <id> bash
exit
docker stop <id>
docker rm <id>
docker rmi ubuntu
```

Exp 7 — Dockerfile

WINDOWS

```
mkdir myapp
cd myapp
@'
<html><body><h1>Hello from Docker!</h1><p>This web page is running inside a Docker container.</p></body>
</html>
'@ | Set-Content -Path index.html -Encoding ascii
@'
FROM nginx
COPY index.html /usr/share/nginx/html
'@ | Set-Content -Path Dockerfile -Encoding ascii
```

LINUX

```
mkdir myapp && cd myapp
cat > index.html <<'EOF'
<html><body><h1>Hello from Docker!</h1><p>This web page is running inside a Docker container.</p></body>
</html>
EOF
printf 'FROM nginx\nCOPY index.html /usr/share/nginx/html\n' > Dockerfile
```

BOTH

```
docker build -t myapp .
docker images
docker run -p 8080:80 myapp     # Jenkins on 8080? use 8081:80
# browser: http://localhost:8080 · 2nd terminal: docker ps
```

Exp 8 — Kubernetes (Minikube)

WINDOWS

```
winget install --id Kubernetes.minikube -e
winget install --id Kubernetes.kubectl -e
mkdir k8s-demo
cd k8s-demo
'<html><body><h1>Hello from Kubernetes!</h1></body></html>' | Set-Content -Path index.html -Encoding ascii
@'
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
'@ | Set-Content -Path Dockerfile -Encoding ascii
```

LINUX (normal user, not root)

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
sudo install minikube-linux-amd64 /usr/local/bin/minikube
curl -LO "https://dl.k8s.io/release/$(curl -L -s
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
sudo install -m 0755 kubectl /usr/local/bin/kubectl
mkdir k8s-demo && cd k8s-demo
echo '<html><body><h1>Hello from Kubernetes!</h1></body></html>' > index.html
printf 'FROM nginx:alpine\nCOPY index.html /usr/share/nginx/html/index.html\nEXPOSE 80\n' > Dockerfile
```

BOTH

```
minikube start --driver=docker
kubectl get nodes
docker build -t k8s-demo:v1 .
docker images k8s-demo
minikube image load k8s-demo:v1
kubectl create deployment k8s-demo --image=k8s-demo:v1 --replicas=2
kubectl get deployments
kubectl get pods
kubectl expose deployment k8s-demo --type=NodePort --port=80
kubectl get services
minikube service k8s-demo
```

Exp 9 — Puppet (Ubuntu 24.04 · Windows = WSL Ubuntu)

```
# Windows only, admin PowerShell: wsl --install -d Ubuntu-24.04
sudo apt update
sudo apt install -y puppet-agent
puppet --version
sudo apt install -y puppetserver
sudo systemctl start puppetserver
sudo systemctl status puppetserver
sudo puppet config print server --section agent
echo "127.0.0.1 puppet" | sudo tee -a /etc/hosts
getent hosts puppet
sudo ss -ltnp | grep 8140
printf "file { '/tmp/puppet-demo':\n  ensure => directory,\n}\n" > first.pp
sudo puppet apply first.pp
ls -ld /tmp/puppet-demo
sudo puppet apply first.pp
sudo puppet agent --test
```

Exp 10 — Puppet blocks (same Ubuntu)

```
mkdir -p ~/puppet-blocks-lab/modules/webdemo/manifests
cd ~/puppet-blocks-lab
cat > modules/webdemo/manifests/init.pp <<'EOF'
class webdemo {
  file { '/tmp/puppet-blocks-demo.txt':
    ensure  => file,
    content => upcase('puppet blocks demonstration'),
  }
}
EOF
echo "include webdemo" > site.pp
sudo puppet apply --modulepath=modules site.pp
cat /tmp/puppet-blocks-demo.txt
sudo puppet apply --modulepath=modules site.pp
```